

Electromagnetic Theory (EE3005)

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Course Instructor(s)

1. Mohsin Yousuf (Course Moderator)
2. Dr. Huzaifa Rauf

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Roll No

Section

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1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q1: In a region of free space, the electric flux density $\mathbf{D}$ is defined by: [20 marks]

$$\mathbf{D} = \begin{cases} \rho_0(z + 2d) \mathbf{a}_z & \text{C/m}^2 \quad \text{for } -2d \leq z \leq 0, \\ -\rho_0(z - 2d) \mathbf{a}_z & \text{C/m}^2 \quad \text{for } 0 \leq z \leq 2d, \\ 0 & \text{elsewhere.} \end{cases}$$

(a) Define Maxwell's first equation (both in point and integral form) and find the volume charge density ρ_v everywhere. [5]

(b) Determine the electric flux ψ passing through the surface along $+\mathbf{a}_z$, defined by $z = 0, -a \leq x \leq a, -b \leq y \leq b$. [5]

(c) Formulate for the total charge enclosed within the following regions: [10]

i) Region 1:

$$-a \leq x \leq a,$$

$$-b \leq y \leq b,$$

$$-d \leq z \leq d$$

call it $Q_{\text{enc},1}$

6) Region 2.

$$-a \leq x \leq a.$$

$$-b \leq y \leq b.$$

$$0 \leq z \leq 2d$$

call it Q_{enc2} .

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q2: Answer the following:

[20 marks]

✓ (a) Define electric dipole and dipole moment $\mathbf{p}$ with the help of diagram.

[4]

✓ (b) A dipole of moment $\mathbf{p} = 3\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z$ [nC · m] is located at the origin in free space. Find V at $P(r = 3.5 \text{ m}, \theta = 30^\circ, \phi = 60^\circ)$.

[6]

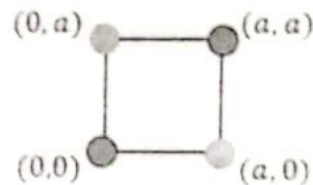
Hint, $V = \frac{\mathbf{p} \cdot \mathbf{a}_r}{4\pi\epsilon_0 r^2}$

(c) Four point-charges $Q_1 = +2q, Q_2 = -q, Q_3 = -q$, and $Q_4 = +2q$ are located

[10]

at $(0, 0), (a, 0), (0, a)$, and (a, a) respectively in yz -plane. Formulate for the potential energy stored in this point charge constellation.

Hint, $W_E = \frac{1}{2} \sum Q_i V_i$



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